

Projected Daily Fire Weather Index Methodology

Degree Day

September 2026

1 Overview

The purpose of this workflow is to estimate how daily fire-weather conditions may change under future climate scenarios. Its principal outputs are daily fields of the Canadian Forest Fire Weather Index System (CFFWIS), including the Fire Weather Index (FWI) and its five component indices, together with annual summary indicators derived from those daily fields.

Daily FWI cannot be calculated reliably from annual or monthly climate projections. It is a sequential index system: today's moisture codes depend on the preceding weather and on the previous day's state. The calculation therefore requires a continuous daily record of temperature, relative humidity, precipitation, and wind speed. Errors in any of these inputs can accumulate or interact nonlinearly in the resulting fire-weather indices.

Raw global climate-model output is not suitable for direct local FWI calculation. Models have systematic biases in the distributions and seasonality of the required variables, use different grids and calendars, and resolve the atmosphere at a much coarser scale than is useful for local fire-weather assessment. In particular, precipitation occurrence and intensity, near-surface humidity, wind, and temperature extremes can differ materially from the historical climate represented by reanalysis data.

The workflow addresses these limitations by placing all models on a common daily 1° grid, adjusting their historical distributions against a reanalysis reference while retaining simulated climate-change signals, and statistically downscaling the adjusted fields to 0.1°. Quality control is applied throughout before the daily CFFWIS indices are calculated with xclim. The result is a consistent set of historical and scenario-based future daily FWI estimates designed for comparing changes across places, years, models, and emissions pathways.

The production workflow processes four variables:

- near-surface air temperature (**tas**);
- near-surface relative humidity (**hurs**);
- precipitation (**pr**); and
- near-surface wind speed (**sfcWind**).

These are the meteorological inputs required to calculate the Canadian Forest Fire Weather Index System (CFFWIS). Other variables, including **prsnratio**, are not downscaled in this production workflow.

2 Climate-model selection

The ensemble uses the five primary climate models selected for ISIMIP3b. ISIMIP selected these models based on data availability, historical performance, process representation, structural independence, and climate sensitivity [9, 6]. Together they include lower- and higher-sensitivity models and therefore sample a broad range of possible climate responses.

Five models capture much of the CMIP6 warming range

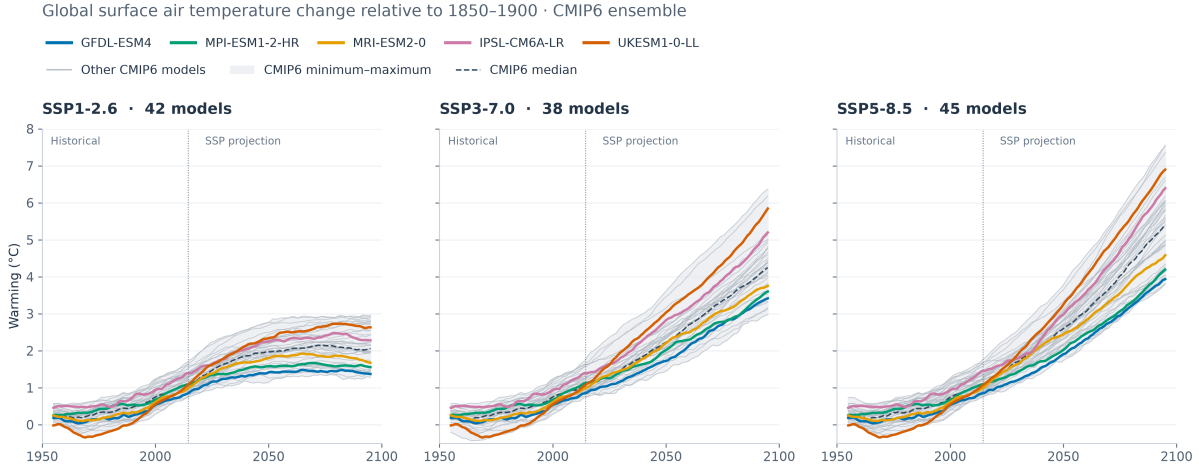


Figure 1: Global temperature trajectories illustrating how the five selected models sample much of the broader CMIP6 warming range under lower-, medium-, and high-forcing pathways. The highlighted models are intended to represent climate-response diversity rather than a probability distribution.

Table 1: Selected models and projection pathways currently available in the Sailfish archive. A dash indicates that the pathway is not currently present.

Model	Sensitivity	SSP1-2.6	SSP2-4.5	SSP3-7.0	SSP5-8.5
GFDL-ESM4	Lower	Yes	Yes	Yes	Yes
MPI-ESM1-2-HR	Lower	Yes	Yes	Yes	Yes
MRI-ESM2-0	Lower	Yes	Yes	Yes	Yes
IPSL-CM6A-LR	Higher	Yes	Yes	—	—
UKESM1-0-LL	Higher	Yes	Yes	—	—

The scenarios range from lower to very high future forcing, as summarized in Table 2. SSP5-8.5 is treated as a high-end stress test, not as a forecast or a “business as usual” scenario. CMIP6 and ScenarioMIP are described by Eyring et al. [5] and O’Neill et al. [12].

3 Input standardization

The raw model data are organized into two periods: historical (1989–2014) and projection (2015–2100). Before bias adjustment, all four variables are standardized to a regular global 1° grid, a complete 365-day no-leap calendar, consistent names and units, and physically valid ranges.

Table 2: SSP pathways represented in the project archive.

Pathway	Interpretation in this workflow	Models available
SSP1-2.6	Lower-forcing pathway	5 of 5
SSP2-4.5	Intermediate-forcing pathway	5 of 5
SSP3-7.0	Medium-high forcing with distinctive aerosol and land-use assumptions	3 of 5
SSP5-8.5	Very-high-forcing pathway used as a high-end stress test	3 of 5

Temperature, humidity, and wind are interpolated to the common grid. Precipitation is conservatively remapped to better preserve spatial totals. Calendar conversion includes specific handling of leap days and 360-day model calendars. Each standardized dataset is checked for complete dates, duplicate or missing time steps, coordinate errors, unexpected dimensions, invalid units, infinities, implausible values, and incomplete active time series.

4 Reference data

The observational reference covers 1993–2014 on nested 1° and 0.1° grids. It is based primarily on ERA5-Land local-noon daily weather [11].

Some mapped coastal land cells are not represented by native ERA5-Land. The reference therefore applies a consistent ERA5-Land coastline repair and uses supplementary regular ERA5 data where needed. Regular ERA5 is bilinearly interpolated from 0.25° to 0.1° and is incorporated only after available ERA5-Land data and the coastline repair [7].

A common land-support mask is used for all four variables. The 0.1° reference supplies the fine-scale spatial pattern, while an area-weighted 1° version supplies the training target for bias adjustment.

This differs from official ISIMIP3b, which used W5E5 at 0.5° resolution. The official configuration is documented by Lange [9].

5 Bias adjustment

Bias adjustment is trained independently for each model, variable, and 1° grid cell using the overlap between the historical model data and the 1993–2014 reference. The workflow uses trend-aware methods from xsdba:

- temperature uses additive detrended quantile mapping;
- relative humidity uses bounded quantile-delta mapping;
- precipitation uses multiplicative quantile-delta mapping with explicit treatment of dry days; and
- wind speed uses multiplicative quantile-delta mapping.

The QDM approach follows Cannon et al. [3]. Conventional quantile mapping can alter a model’s projected trends, particularly in precipitation extremes. QDM instead adjusts the historical distribution toward the reference while explicitly preserving the model-projected change at corresponding quantiles. A moving seasonal window represents the annual cycle without introducing abrupt month-boundary transitions.

The adjustment is univariate: each weather variable is treated separately. The inter-variable MBCn

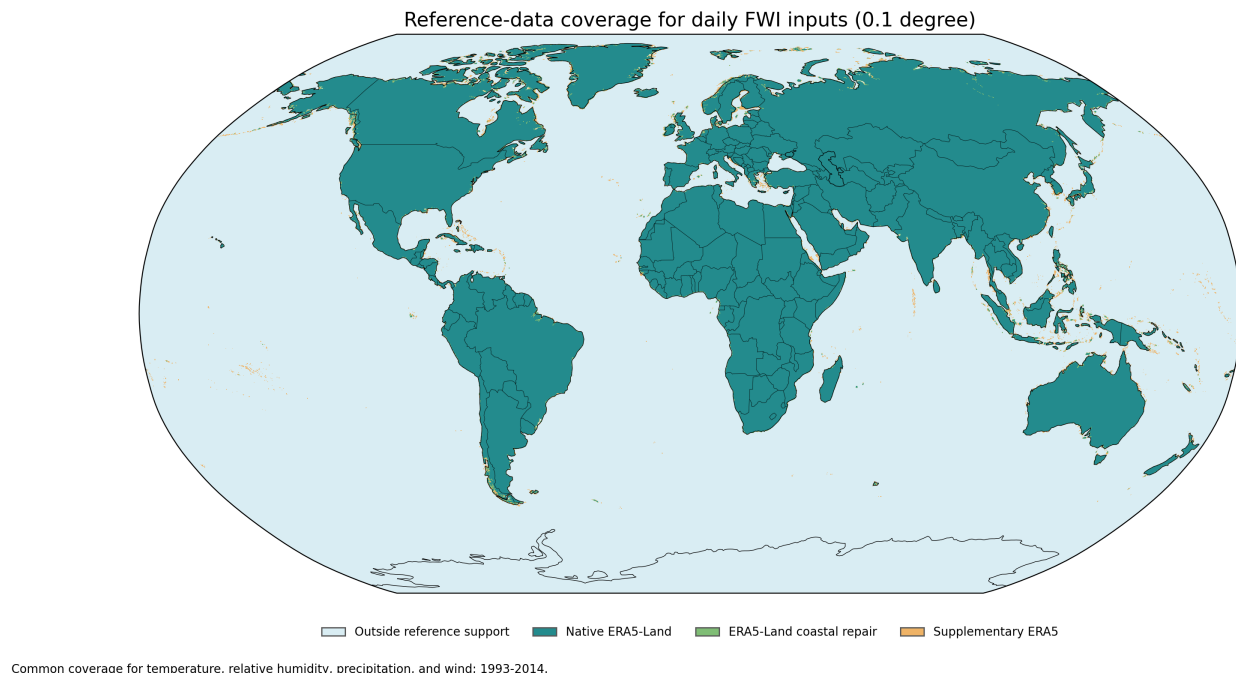


Figure 2: Coverage and source composition of the 0.1° daily reference used for temperature, relative humidity, precipitation, and wind speed. Native ERA5-Land is retained wherever available; mapped coastal land is completed by the common ERA5-Land repair and supplementary regular ERA5 data.

bias-adjustment stage is not used. ISIMIP3b also omitted that optional stage after it produced overfitting and spatial-coherence problems [9]. Trained relationships can be cached and reused for historical and future periods without changing the resulting values.

6 Spatial downscaling

After bias adjustment, each 1° field is downscaled separately to the 0.1° reference grid using MBCnSD. At a high level, the method:

1. interpolates the adjusted model field to the fine grid;
2. learns the historical distribution among fine cells within each coarse parent cell;
3. adjusts the fine-grid values toward that spatial distribution; and
4. preserves the area-weighted climate-model signal of the parent cell.

MBCnSD is based on multivariate quantile mapping, but its vector components in this application are fine spatial cells, not different weather variables [4, 8]. The calculation is divided into restartable geographic tiles. Each tile reads surrounding spatial context before calculation, preventing processing-region boundaries from creating artificial edges.

A 0.1° output grid does not mean that the GCM dynamically resolves climate at 0.1° . Fine-scale patterns are statistically inferred from the historical reference, while the future signal still comes from the coarser model. Some 1° parent-cell structure can remain, particularly in daily precipitation and derived FWI fields.

California climatology: 1° to 0.1°

ACCESS-CM2 · SSP2-4.5 · Annual means, 2066–2095

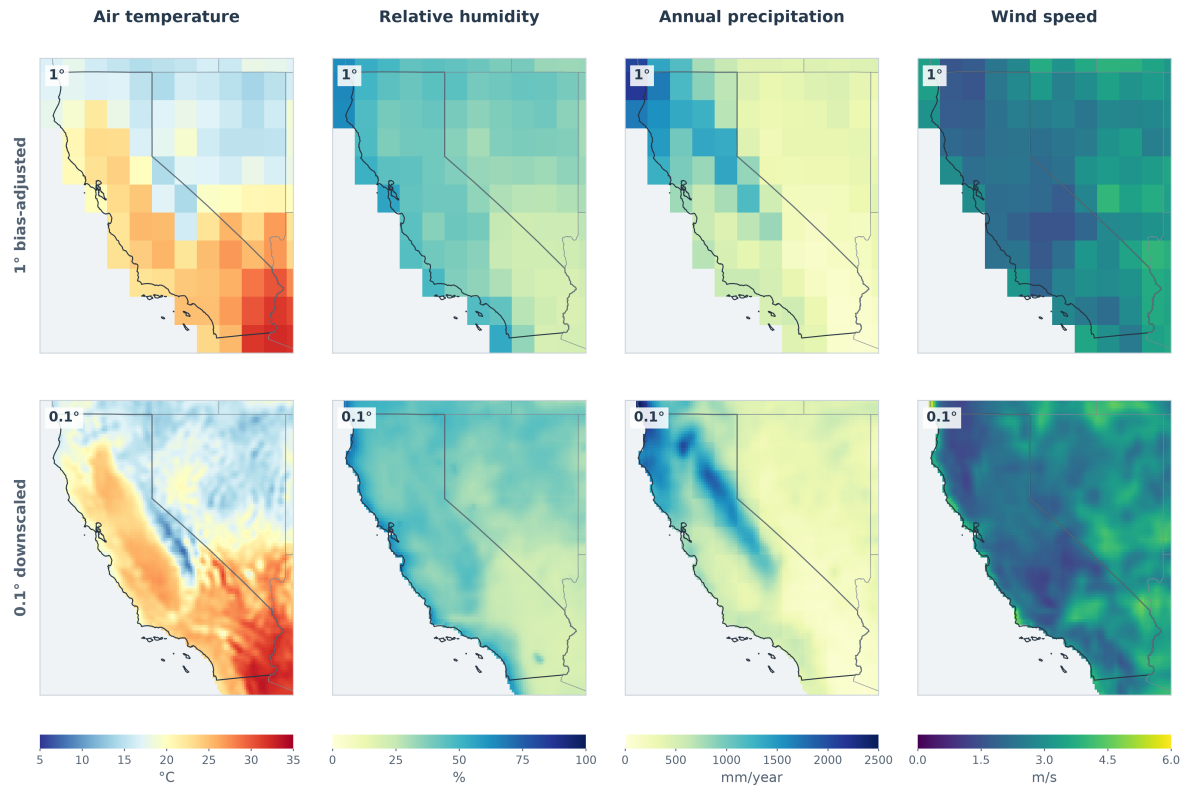


Figure 3: Illustrative California climatologies before and after spatial downscaling from 1° to 0.1° for ACCESS-CM2 under SSP2-4.5, averaged over 2066–2095. The finer fields recover reference-scale spatial patterns while retaining the climate-model signal at the coarser scale.

7 Quality control and storage

Quality control is performed at every major stage. It checks chronology, calendar and coordinates, common coverage among variables, physical ranges, missing supported land cells, spatial discontinuities, preservation of the adjusted 1° signal, and numerical accuracy after storage.

Global maps and regional views are inspected in addition to machine-readable tests. This helps distinguish processing-tile seams from spatial structure inherited from the parent climate-model grid. Extreme precipitation is capped at $3,000 \text{ mm day}^{-1}$, following a control used by the Climate Impact Lab downscaleCMIP6 workflow [14].

Final daily products are stored as compressed, scaled `int16` Zarr datasets. Scale factors and offsets permit transparent decoding to physical values. A reserved integer code distinguishes missing data from valid zeroes.

8 Canadian Forest Fire Weather Index System

All six daily CFFWIS components are calculated with xclim’s `xclim.indices.fire.cffwis_indices` routine [2]. The workflow does not reimplement the Canadian FWI equations. It supplies the downscaled inputs, maintains the calculation through time, applies the common land mask, and validates and stores the outputs.

The daily outputs are the Fine Fuel Moisture Code (`ffmc`), Duff Moisture Code (`dmc`), Drought Code (`dc`), Initial Spread Index (`isi`), Build Up Index (`bui`), and Fire Weather Index (`fwi`). The calculation uses temperature, relative humidity, precipitation, wind speed, and latitude. It includes a temperature-defined fire season and Drought Code overwintering. The system is described by Van Wagner [13].

8.1 Spin-up, boundary continuity, and restarts

CFFWIS is stateful: the moisture codes on one day depend on their values on the preceding day. At the beginning of a calculation, xclim applies the standard start-up values of 85 for FFMC, 6 for DMC, and 15 for DC. The WF93 season method then identifies seasonal start-up and shutdown, and DC overwintering carries information from one fire season into the next.

The historical calculation begins on 1 January 1989. No separate historical spin-up file is published. The years before the 1995–2014 reference interval provide six complete years of model weather and CFFWIS evolution before the historical thresholds and summary statistics are evaluated. This reduces sensitivity of the reported reference-period indicators to the initial values, although the earliest daily historical output should still be regarded as initialization-affected.

For each model-scenario pathway, the projection calculation prepends the 1989–2014 historical meteorology before calling xclim. CFFWIS state therefore evolves continuously across 31 December 2014 and 1 January 2015, including in the Southern Hemisphere and other locations where the fire season may be active at the archive boundary. Only the 2015–2100 portion is written to the projection store. This preserves the convenient historical and projection archive layout without reinitializing FFMC, DMC, DC, fire-season state, or DC overwintering at the boundary.

Computational restart files serve a different purpose. The global calculation is divided into spatial

tiles, and a versioned success marker is written only after all six indices for a tile have been stored and checked. A completed tile is skipped when a job restarts; an interrupted tile is recomputed over its full requested time interval. These markers are workflow checkpoints, not saved FFMFC, DMC, DC, season, or overwintering states. The implementation does not resume a tile from an intermediate date. The checkpoint version changes when continuity semantics change, preventing outputs made with an older initialization scheme from being silently accepted.

Four annual indicators are derived from the daily xclim output:

fwixx
annual maximum FWI;
fwixd
annual count of days above the local historical 95th-percentile FWI threshold;
fwils
annual count of days above the local historical FWI midrange; and
fwisa
annual maximum 90-day mean FWI.

The local thresholds are calculated from 1995–2014 and then held constant for historical and future years. These indicators follow the concepts used by Abatzoglou et al. [1].

FWI describes weather conditions associated with potential fire behaviour. It does not predict ignition, burned area, structure loss, smoke, or suppression difficulty. Those outcomes also depend on fuels, vegetation, land management, topography, human activity, and exposure.

9 Interpretation and limitations

The product is based on climate scenarios rather than forecasts. Statistical adjustment cannot correct errors in the model’s underlying climate response, and the fine-scale relationships observed during 1993–2014 are assumed to remain applicable in future climates. The four variables are adjusted independently, and the 0.1° grid contains statistically inferred detail rather than new high-resolution climate physics.

Reference uncertainty and the differing time semantics of supplementary regular ERA5 data affect some coastal locations. Model and scenario uncertainty must therefore be retained in downstream analysis. Results should be reported across models and scenarios rather than as a single deterministic future. The approach draws on ISIMIP3BASD methods, but the resulting dataset is not an official ISIMIP3b forcing product or a bit-for-bit reproduction of the archived ISIMIP3BASD software [10]. Source-model licenses, institutional credits, software versions, reference-data citations, and processing provenance must accompany published or redistributed products.

References

- [1] Abatzoglou, J. T., Williams, A. P., and Barbero, R. (2019). Global emergence of anthropogenic climate change in fire weather indices. *Geophysical Research Letters*, 46, 326–336. <https://doi.org/10.1029/2018GL080959>.

Daily FWI in observed and future water years

Los Angeles–San Fernando study area · three high-peak future water years from ACCESS-CM2 under SSP2-4.5

Each row is one complete water year (October–September); all panels use the same FWI scale.

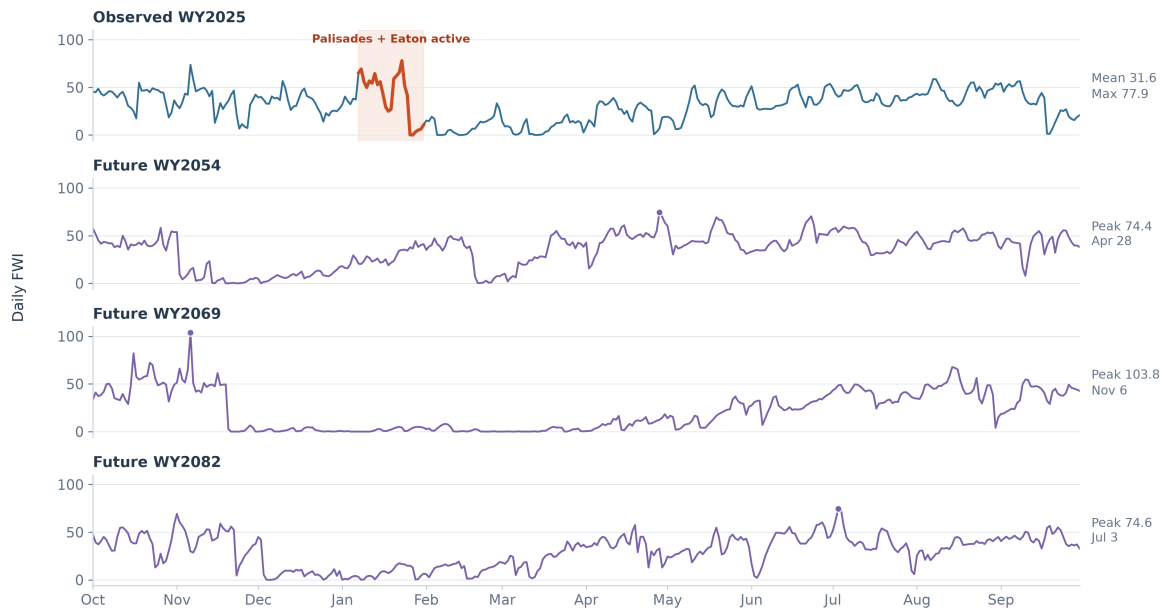


Figure 4: Example daily FWI sequences for the Los Angeles–San Fernando study area. The observed 2025 water year is shown alongside three ACCESS-CM2 SSP2-4.5 water years selected for high seasonal peaks. Individual future years are scenario realizations used to examine plausible daily conditions; they are not forecasts for those calendar years.

- [2] Bourgault, P., et al. (2023). xclim: xarray-based climate data analytics. *Journal of Open Source Software*, 8, 5415. <https://doi.org/10.21105/joss.05415>.
- [3] Cannon, A. J., Sobie, S. R., and Murdock, T. Q. (2015). Bias correction of GCM precipitation by quantile mapping: How well do methods preserve changes in quantiles and extremes? *Journal of Climate*, 28, 6938–6959. <https://doi.org/10.1175/JCLI-D-14-00754.1>.
- [4] Cannon, A. J. (2018). Multivariate quantile mapping bias correction: an N-dimensional probability density function transform for climate model simulations of multiple variables. *Climate Dynamics*, 50, 31–49. <https://doi.org/10.1007/s00382-017-3580-6>.
- [5] Eyring, V., et al. (2016). Overview of the Coupled Model Intercomparison Project Phase 6 experimental design and organization. *Geoscientific Model Development*, 9, 1937–1958. <https://doi.org/10.5194/gmd-9-1937-2016>.
- [6] Frieler, K., et al. (2026). Scenario set-up and the new CMIP6-based climate-related forcings provided within ISIMIP3b. *Geoscientific Model Development*, 19, 4095–4146. <https://doi.org/10.5194/gmd-19-4095-2026>.
- [7] Hersbach, H., et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146, 1999–2049. <https://doi.org/10.1002/qj.3803>.
- [8] Lange, S. (2019). Trend-preserving bias adjustment and statistical downscaling with ISIMIP3BASD. *Geoscientific Model Development*, 12, 3055–3070. <https://doi.org/10.5194/gmd-12-3055-2019>.
- [9] Lange, S. (2021). ISIMIP3b bias adjustment fact sheet. https://www.isimip.org/documents/413/ISIMIP3b_bias_adjustment_fact_sheet_Gnsz7C0.pdf.
- [10] Lange, S. (2022). ISIMIP3BASD v3.0.2 source archive. <https://zenodo.org/records/7151476>.
- [11] Munoz-Sabater, J., et al. (2021). ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data*, 13, 4349–4383. <https://doi.org/10.5194/essd-13-4349-2021>.
- [12] O’Neill, B. C., et al. (2016). The Scenario Model Intercomparison Project for CMIP6. *Geoscientific Model Development*, 9, 3461–3482. <https://doi.org/10.5194/gmd-9-3461-2016>.
- [13] Van Wagner, C. E. (1987). *Development and Structure of the Canadian Forest Fire Weather Index System*. Canadian Forestry Service, Forestry Technical Report 35. https://www.canadawildfire.org/_files/ugd/90df79_f7ee78aa08b54e4a9b691a802dbb30c1.pdf.
- [14] Climate Impact Lab (2024). Global Downscaled Projections for Climate Impacts Research: downscaleCMIP6. <https://github.com/ClimateImpactLab/downscaleCMIP6>.